

15/07/2020

(12)

Integer to Roman

Dry Run

* num = 1994

Values $\Rightarrow \{ \underline{1000}, \underline{900}, \underline{500}, 400, 100, 90, 50$

$, 40, 10, 9, 5, 4, 1 \}$

Symbol $\Rightarrow \{ "M", "CM", "D", "CD",$

"C", "XC", "L", "XL"

"X", "IX", "V", "IV", "I" }

Relation b/w two arr $\Rightarrow$

1000 $\rightarrow$ M

900 $\rightarrow$ CM

500 $\rightarrow$ D

400 $\rightarrow$ CD

100 $\rightarrow$ C

90 $\rightarrow$ XC

50 $\rightarrow$ L

40 $\rightarrow$ XL

10 $\rightarrow$ X

9 $\rightarrow$ IX

5 $\rightarrow$ V

4 $\rightarrow$ IV

1 $\rightarrow$ I

→ logic Part

(for) ↳ We iterate in arr
while ↳ check the values which
is less / equal to num
↳ append the symbol
↳ remove the value
in num.

→ return ans

① num $\rightarrow$ 1994

ans $\rightarrow$ "MCMXC"

$(1994 \geq 1000)$ T

↳ $1994 - 1000 \rightarrow 994$

$(994 \geq 1000)$ (F) $(994 \geq 900)$ T

↳ $994 - 900 \rightarrow 94$

$(94 \geq 1000)$ (F) $(94 \geq 900)$ (F) ... $(94 \geq 90)$ (T)

(F)

(F)

↳ $94 - 90 \rightarrow 4$

$(4 \geq 1000)$

(F)

$(4 \geq 4)$

(T)

↳ $4 - 4 \rightarrow 0$

0

→ While getting 0 we ~~stop~~ end
while ~~but~~ because we are
not having any number ~~get~~
greater / equal to zero.

→ Cool →

String helper (int num) {

int [] values = {

1000, 900, 500, 400,

100, 90, 50, 40,

10, 9, 5, 4, 1

};

String

~~Symbol~~ [] symbol = {

"M", "CM", "D", "CD",

"C", "XC", "L", "XL",

"X", "IX", "V", "IV", "I"

};

String Builder ans a new String Builder;

```
for (int i = 0; i < values.length; i++) {
```

```
    while (num > values[i]) {
```

```
        ans.append (symbol[i]);
```

```
        num -= values[i];
```

```
    }
```

```
    return ans.toString();
```

```
}
```